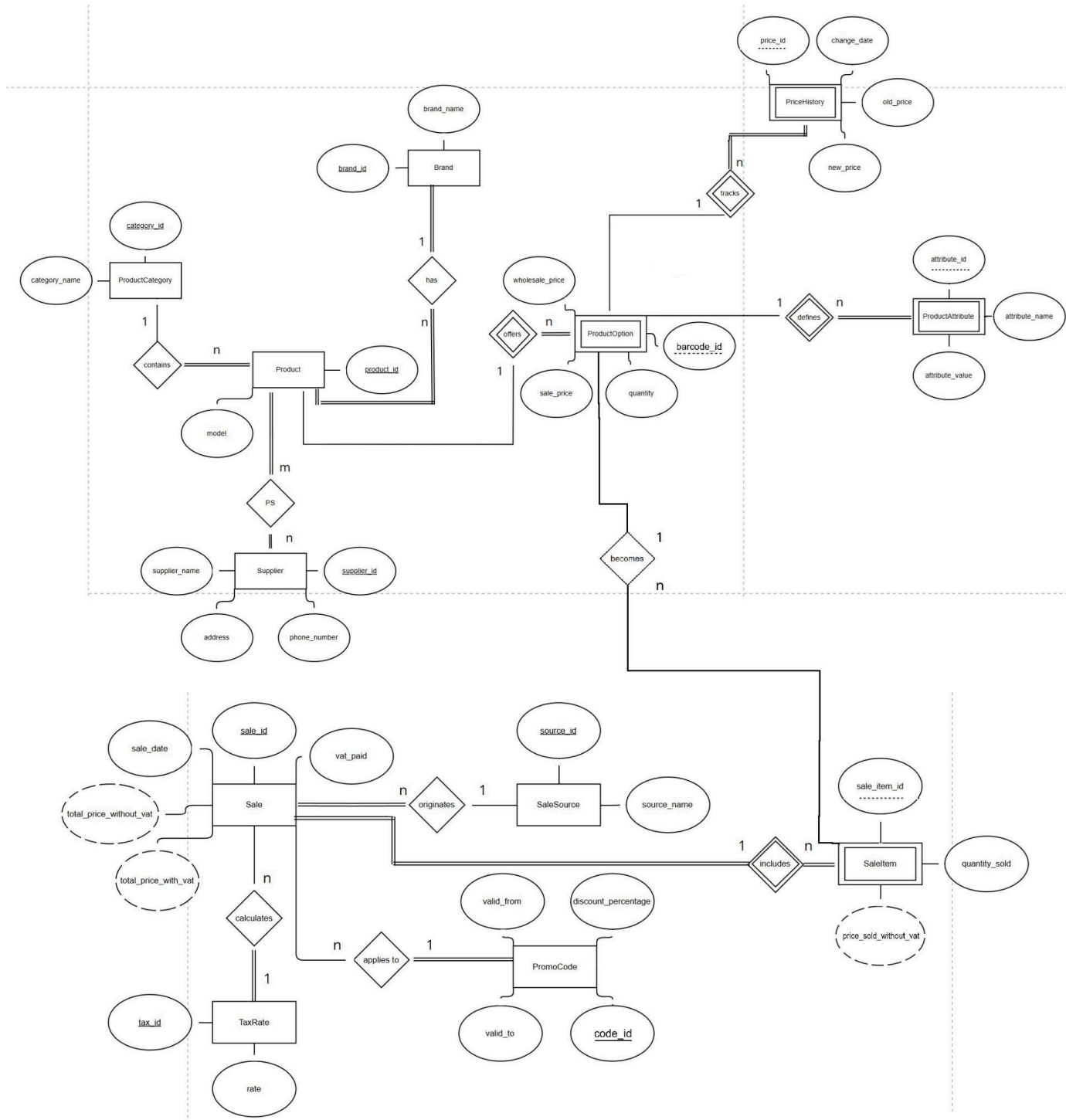
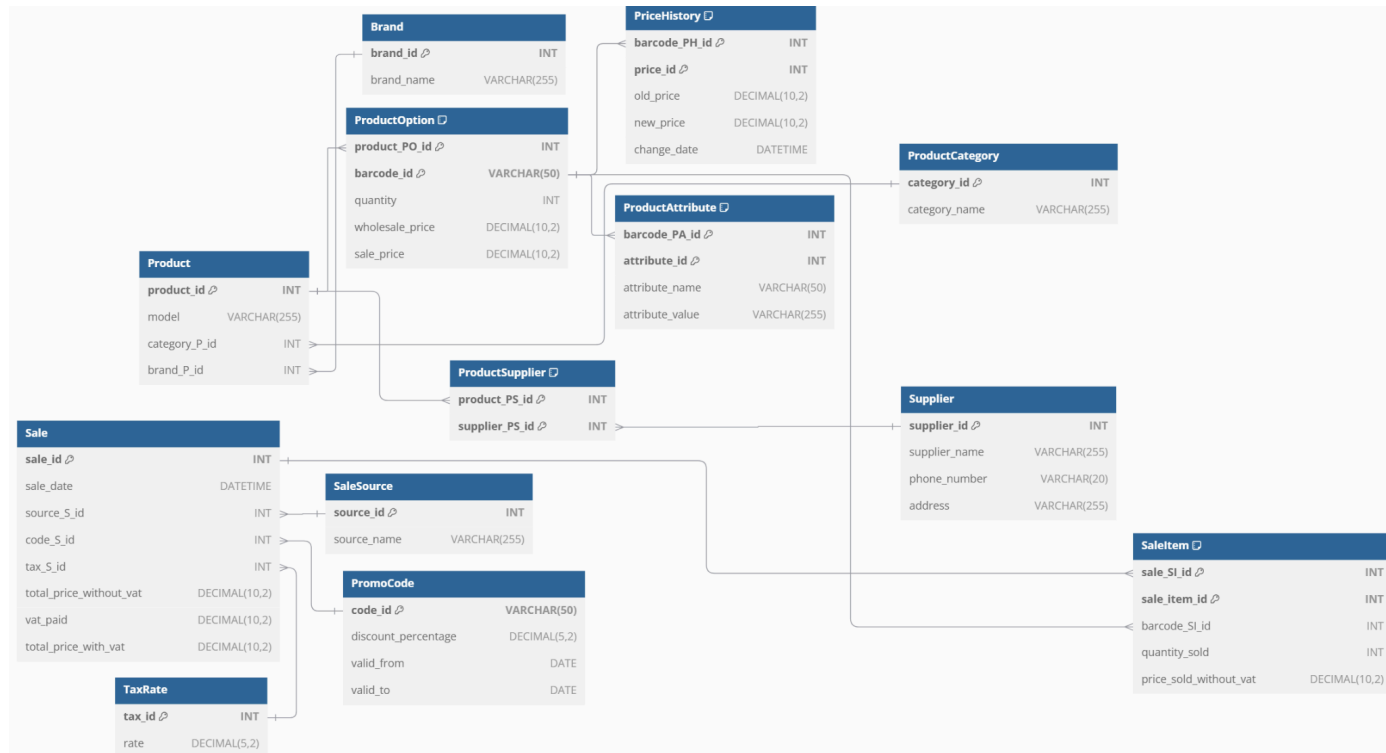


Assignment 1 / Phase 1- Design the ER Diagram and the Relational Model.

1)ER model.



2)Relational model.



Relations.

All relationships are One-to-Many (1:N) except for one: the relationship between Product and Supplier, which is Many-to-Many (N:M). A single Product can be supplied by multiple Suppliers, and a single Supplier can supply multiple products.

Primary Keys:

1. Product: product_id
2. ProductOption: product_PO_id, barcode_id
3. ProductAttribute: barcode_PA_id, attribute_id
4. ProductCategory: category_id
5. PriceHistory: barcode_PH_id, price_id
6. Sale: sale_id
7. SaleItem: sale_SI_id, sale_item_id
8. SaleSource: source_id
9. PromoCode: code_id
10. TaxRate: tax_id
11. ProductSupplier: product_PS_id, supplier_PS_id
12. Supplier: supplier_id
13. Brand: brand_id

Foreign Keys:

1. Product.

category_P_id → ProductCategory.category_id

brand_P_id → Brand.brand_id

2. ProductOption.

product_PO_id → Product.product_id

3. ProductAttribute.

barcode_PA_id → ProductOption.barcode_id

4. PriceHistory.

barcode_PH_id → ProductOption.barcode_id

5. Sale.

source_S_id → SaleSource.source_id

code_S_id → PromoCode.code_id (can be NULL)

tax_S_id → TaxRate.tax_id

6. SaleItem.

sale_SI_id → Sale.sale_id

barcode_SI_id → ProductOption.barcode_id

7. ProductSupplier.

product_PS_id → Product.product_id

supplier_PS_id → Supplier.supplier_id

Constraints:

ProductOption(barcode must be unique).

PromoCode(code must be unique).

Sale(promo_S_id can be NULL).

Supplier(phone_number and address can be NULL).

3) So, at the center we have the main table **Product**, which represents our products. Each product has a model and belongs to a specific **brand** and **category** (for example, Samsung as a brand and "phones" as a category).

Each product can have different variations, which are stored in the **ProductOption** table. Each option has its own barcode, quantity, purchase price and selling price. For instance, if we have an iPhone, it may come in different colors and storage capacities. All these variations are recorded in the **ProductAttribute** table.

When a product's price changes, it is recorded in the **PriceHistory** table. This table stores the old price, the new price, and the date of the change making it convenient to track price changes over time.

At the bottom, we have the **Sale** table, which represents our sales. Each sale has a date and may be associated with a **promo code**(if it was used). The **PromoCode** table stores promo codes, their discount amounts and validity periods.

The **SaleItem** table records the items within a sale. If a customer purchases three different products, there will be one record in the **Sale** table(representing the sale itself) and three records in **SaleItem**(representing the purchased products).

Lastly, we have the **Supplier** table, which stores information about our suppliers, including their name, phone number and address. The **ProductSupplier** table links products to suppliers because a single product can be supplied by multiple companies.